ModelIt!

Teacher's Guide

Plate Tectonics: Predicting Where Earth Will Crack Next

Using Models to Forecast Earthquakes and Volcanic Activity

Quick Reference

Grade Level	Grade 06L2
NGSS Standard	MS-ESS2-2, MS-ESS2-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-ESS2-2, MS-ESS2-3

Construct an explanation based on evidence for how geoscience processes have changed Earth's surface at varying time and spatial scales. Analyze and interpret data on the distribution of fossils and rocks, continental shapes, and seafloor structures to provide evidence of the past plate motions.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how mantle convection, boundary type, and tectonic stress interact to produce patterns of earthquakes and volcanic activity.
Disciplinary Core Idea	**ESS2.B Plate Tectonics and Large-Scale System Interactions** — Tectonic plates are the top parts of giant convection cells that bring hot material up from Earth's interior and send cooled material back down. Plate interactions cause earthquakes, volcanic activity, and mountain building.
Crosscutting Concept	**Stability and Change** — Students explore how plate boundaries appear stable over human timescales but are zones of constant stress accumulation and sudden release, creating cyclical patterns of geological activity.

CAST Testing Alignment

MULTIPLE CHOICE:

Scientists observe that a transform fault has been quiet for 150 years while a similar fault nearby has frequent small earthquakes. Based on tectonic stress models, which fault poses a greater risk of a large earthquake, and why?

CONSTRUCTED RESPONSE:

Using your plate tectonics model, explain why Japan experiences both earthquakes AND volcanic eruptions while the San Andreas Fault area in California has earthquakes but very few volcanoes. Reference boundary types in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Plate tectonics is the unifying theory of geology, explaining earthquakes, volcanoes, mountain building, and even the distribution of fossils. By modeling the stress dynamics at different boundary types, students can predict real-world geological patterns.

Mantle Convection Drives Everything

Earth's interior is incredibly hot (over 5,000 degrees C at the core). This heat creates convection currents in the mantle: hot rock rises, spreads laterally, cools, and sinks. These currents drag tectonic plates along at rates of 2-15 cm per year. Though slow by human standards, these movements reshape continents over millions of years.

Three Boundary Types, Three Hazard Profiles

Convergent boundaries (plates collide) create mountains and subduction zones where one plate dives under another, melting and fueling volcanoes. Divergent boundaries (plates separate) create mid-ocean ridges and rift valleys. Transform boundaries (plates slide past) create earthquake faults. Each type produces a distinct combination of hazards.

The Stress-Release Cycle

At plate boundaries, friction locks plates together while convection keeps pushing. Stress accumulates over years or decades. When stress exceeds the frictional strength of the rock, it fails suddenly, releasing energy as an earthquake. After the quake, stress drops and begins rebuilding. This creates a cyclical pattern that is predictable in average frequency but unpredictable in exact timing.

Prediction vs. Forecasting

Scientists cannot predict the exact date, time, and location of an earthquake. But they CAN forecast probabilities: the USGS estimates a 72% chance of a magnitude 6.7+ earthquake in the San Francisco Bay Area before 2043. This is based on historical patterns, stress measurements, and understanding of the recurrence intervals at specific faults.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify mantle convection speed and plate boundary type as external inputs, and tectonic stress, earthquake frequency, and volcanic activity as internal responses of the system.

E **Establish Relationships**

Students determine that mantle convection positively drives tectonic stress, stress positively drives earthquake frequency, but earthquakes negatively feed back to reduce stress (negative feedback). Boundary type determines whether volcanic activity occurs.

V **Visualize & Model**

Students build a model showing how convection drives stress accumulation, which is periodically released by earthquakes, creating cyclical patterns. Different boundary types produce different hazard combinations.

E **Evaluate Outcomes**

Students run fast convergent vs. slow transform scenarios to compare hazard profiles and observe the stress-release cycle in the simulation graphs.

R **Revise & Extend**

Students add plate thickness or magma viscosity to explore why some convergent boundaries produce explosive volcanoes while others produce gentle eruptions.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Right now, the ground under you is moving. You cannot feel it, but the forces building beneath your feet could release in seconds."*

Watch For

Show a global earthquake map with the Ring of Fire highlighted. Let students notice the pattern.

Slide 2: Learning Objectives

3 min

Say: *"Today we are going beyond plate tectonics to model the STRESS system that controls when and where Earth cracks."*

Watch For

Have students read objectives. Pre-teach 'tectonic stress' and 'convergent/divergent/transform.'

Slide 3: Big Question

3 min

Say: *"We know California will have a big earthquake. We know the fault that will do it. So why can't we say when?"*

Watch For

Think-pair-share: What makes earthquake timing so hard to predict?

Slide 4: LEVER Framework

2 min

Say: *"Our model has a hidden feedback loop: earthquakes release stress, which REDUCES earthquake frequency. Until stress builds again."*

Watch For

Preview the negative feedback concept. Draw the cycle on the board.

Slide 5: Activity 1: Components

8 min

Say: *"What drives the system? What responds? Notice we have TWO external inputs this time: speed AND boundary type."*

Watch For

Guide sorting. Explain why boundary type is external (it is a geological given, not something the system changes).

Slide 6: Activity 2: Connections

8 min

Say: *"The critical relationship: earthquakes RELEASE stress. So more earthquakes at a boundary temporarily REDUCE the stress driving them."*

Watch For

Demonstrate with a spring: compress (stress builds), release (earthquake), spring returns to normal (stress resets).

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's compare a fast convergent boundary to a slow transform boundary. Which is more dangerous?"*

Watch For

Students predict first. Run both scenarios. Compare earthquake frequency AND volcanic activity side by side.

Slide 8: Discoveries

5 min

Say: *"Now you understand why we can forecast earthquake ZONES but not exact timing. The cycle is real but the timing is chaotic."*

Watch For

Show USGS earthquake probability forecasts. Discuss how models inform preparedness even without exact prediction.

Slide 9: STEM Challenge

12 min

Say: *"Your city sits near TWO fault types. Design a warning system that handles both earthquake and volcanic threats."*

Watch For

Distribute maps and materials. Groups design integrated warning systems using model evidence.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Mantle Convection Speed, Plate Boundary Type
- **INTERNAL:** Tectonic Stress, Earthquake Frequency, Volcanic Activity

Relationships

- **Mantle Convection Speed to Tectonic Stress:** POSITIVE (+) — Faster mantle convection pushes plates harder, accumulating more stress at boundaries where plates interact. More convection energy means more force building up.
- **Tectonic Stress to Earthquake Frequency:** POSITIVE (+) — Higher accumulated stress means more stored energy that must be released. This increases both the likelihood and frequency of seismic events at the boundary.
- **Plate Boundary Type to Volcanic Activity:** POSITIVE/NEGATIVE (+/-) — Convergent boundaries produce volcanoes because subducting plates melt and create magma. Transform boundaries produce almost no volcanic activity because plates slide horizontally without subduction. Boundary type determines the volcanic hazard profile.
- **Earthquake Frequency to Tectonic Stress:** NEGATIVE (-) — Earthquakes release accumulated stress, temporarily reducing the pressure at the boundary. This creates a negative feedback cycle: stress builds, earthquake releases, stress drops, then rebuilds over time.
- **Mantle Convection Speed to Volcanic Activity:** POSITIVE (+) — Faster convection brings more heat to the surface, increasing magma production and volcanic activity at convergent and divergent boundaries.

Key Discoveries

- Faster mantle convection increases tectonic stress at boundaries, leading to more frequent geological events
- Convergent boundaries produce BOTH earthquakes and volcanoes because subducting plates melt and create magma
- Transform boundaries produce earthquakes but almost NO volcanoes because plates slide horizontally without subduction
- Earthquakes release accumulated stress, creating a negative feedback cycle: stress builds, quake releases, stress rebuilds
- The stress-release cycle explains why earthquake patterns are somewhat cyclical but never perfectly predictable

Multiple Choice Post-Assessment Answer Key

1. Why do convergent plate boundaries produce volcanoes but transform boundaries do not?

Correct: B) At convergent boundaries, one plate subducts into the mantle and melts, creating magma that rises as volcanoes. Transform boundaries have no subduction.

Correct! At convergent boundaries, one plate dives beneath the other (subduction) into the hot mantle, where it partially melts. This magma rises through the overlying plate and erupts as volcanoes. At transform boundaries, plates slide horizontally past each other with no subduction, so no melting occurs and no magma is produced.

2. In the plate tectonics model, tectonic stress builds and then drops repeatedly. What creates this cyclical pattern?

Correct: C) Earthquakes release accumulated stress (negative feedback), then convection rebuilds the stress, creating a repeating cycle

Correct! This is a negative feedback cycle. Mantle convection continuously pushes plates, building stress at boundaries. When stress exceeds the friction holding plates together, an earthquake releases that stored energy. Stress drops, then gradually rebuilds as convection continues pushing. This creates the cyclical build-release pattern.

3. A region on a plate boundary has not experienced an earthquake in 200 years, while a nearby boundary has frequent small earthquakes. Which region poses a greater risk of a large earthquake?

Correct: C) The quiet region, because 200 years of stress accumulation without release means enormous energy is stored

Correct! The quiet region has been accumulating stress for 200 years without release. The model shows that longer periods without earthquakes mean more stored energy, which means a potentially larger earthquake when the fault finally slips. The region with frequent small quakes is regularly releasing small amounts of stress.

4. How does increasing mantle convection speed affect geological activity at a plate boundary?

Correct: C) Faster convection pushes plates harder, building stress faster and increasing earthquake frequency and volcanic activity

Correct! Faster mantle convection drives plates together (or apart, or past each other) with more force. This builds tectonic stress more rapidly, leading to more frequent earthquakes. At convergent boundaries, faster convection also increases heat delivery and magma production, intensifying volcanic activity.

5. Scientists know California will have a major earthquake, but they cannot predict the exact date. What does the plate tectonics model reveal about why precise timing is impossible?

Correct: B) The exact moment when accumulated stress overcomes rock friction depends on countless microscopic variables that are impossible to measure

Correct! The model shows that the stress-release pattern is predictable on average, but the exact moment of failure depends on microscopic conditions in the rock: tiny fractures, water content, mineral composition, and temperature variations. It is like knowing a fraying rope will break but not being able to predict which fiber fails first.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Earthquakes are random events"	Earthquakes are NOT random. They occur at plate boundaries in patterns governed by stress accumulation and release. While we cannot predict exact timing, the locations and average frequencies are well understood and follow the physics of plate tectonics.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"All plate boundaries are the same"	Each boundary type produces distinctly different hazards. Convergent boundaries create mountains, volcanoes, and deep earthquakes. Divergent boundaries create rift valleys and mid-ocean ridges. Transform boundaries create horizontal earthquake faults. Knowing the boundary type tells you WHAT to prepare for.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"After a big earthquake, the area is safe for a long time"	While stress is temporarily reduced after a large earthquake, aftershocks can occur for weeks or months as the surrounding rock adjusts to the stress change. Additionally, releasing stress on one part of a fault can actually INCREASE stress on adjacent sections, potentially triggering earthquakes nearby.